

Today's plan

1. A Brief Review of the Generalized Spectral Theory
2. Application to the Kuramoto Model (Synchronization)
3. Application to the neuronal dynamics (brain wave)

Generalized Spectral Theory and its Applications

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A linear operator $A: X \rightarrow X$
defined on a Banach sp. X .

A spectral set $\sigma(A)$ plays
an important role in many areas.

• When $X = \mathbb{C}^n$ (i.e. A is a matrix)

$$\lambda \in \sigma(A)$$

$$\Leftrightarrow (\lambda - A)u = 0, \quad \exists u \neq 0$$

$$\Leftrightarrow \det(\lambda - A) = 0$$

$\Leftrightarrow$ The inverse

$$(z - A)^{-1} = \frac{(\text{adjugate matrix})}{\det(z - A)}$$

rational func. of z .

has a singularity at $z = \lambda$.

$$\sigma(A) = \{ \text{The set of singularities} \\ \text{of } (z - A)^{-1} \}$$

$(z - A)^{-1}$ is called the
resolvent of A .

X : Banach sp.

$A: X \rightarrow X$, linear op.

Def

The set of singularities of $(z - A)^{-1}$ on $\mathbb{C}$ is called the spectral set $\sigma(A)$.

that consists of ...

eigen values

continuous spectrum

residual spectrum

} difficult

Purpose

Generalize the concept of the spectral set to treat problems involving conti / res. spectra.

The Generalized Spectral

Theory

H. Chiba, Adv. in Math. (2015)

Ex. 1 (Kuramoto Model, 1975)

A model for synchronization.

$$\frac{d\theta_i}{dt} = \omega_i + \frac{K}{N} \sum_{j=1}^N \sin(\theta_j - \theta_i)$$

$(i = 1, \dots, N)$

$$\theta_i \in [0, 2\pi)$$

ω_i : natural frequencies drawn from a distribution $f(\omega)$.

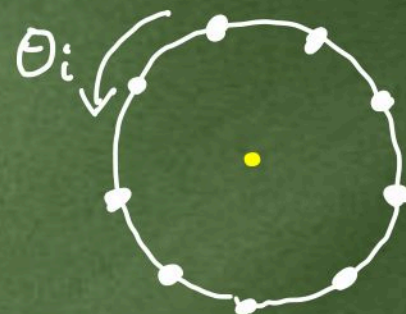
Gaussian etc...

K : coupling strength
(bifurcation parameter)

The order parameter

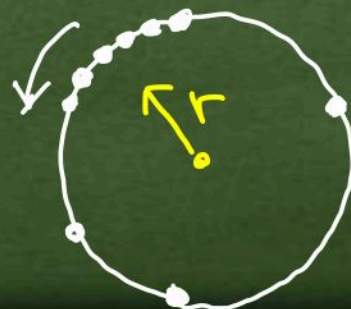
$$r := \frac{1}{N} \sum_{j=1}^N e^{i\theta_j}$$

$\exists$ threshold $K_c > 0$ s.t.



$0 < K < K_c$
de-synchro
 $r \sim 0$

↓ phase transition



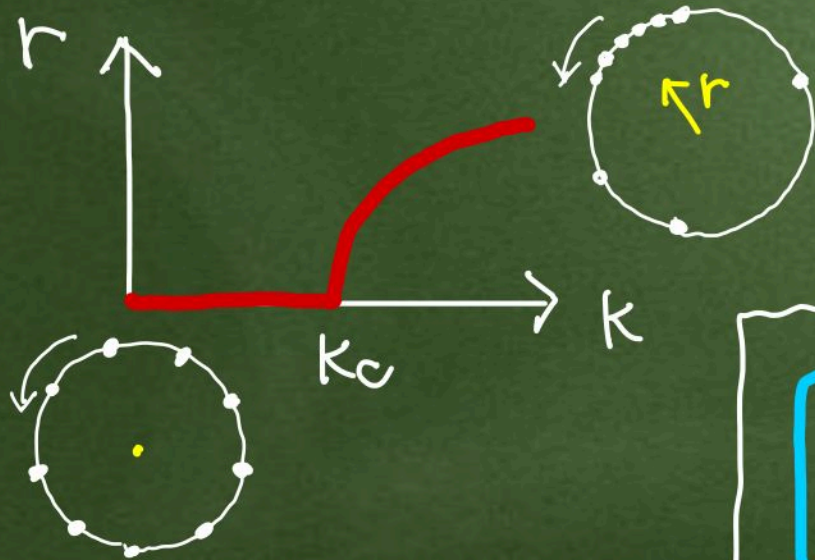
$K_c < K$
synchro
 $r > 0$

Kuramoto Conjecture

$$N \rightarrow \infty$$

(KM becomes a certain PDE) $\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \theta}(\rho \omega) = 0$

A bifurcation diagram of the order parameter is given as

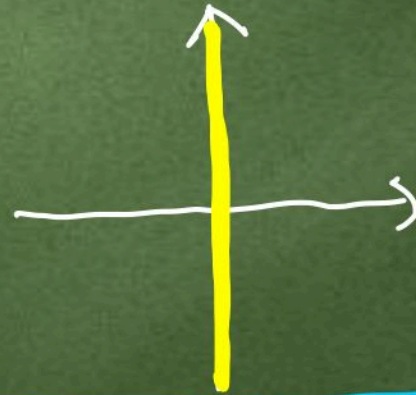


Linearization of the KM around the de-synchro state (uniform dist. on S^1)

Linearized op.

$$A : L^2(\mathbb{R}, g(\omega) d\omega) \rightarrow L^2(\mathbb{R}, g(\omega) d\omega)$$

The spectral set $\sigma(A)$ is $i\mathbb{R}$.

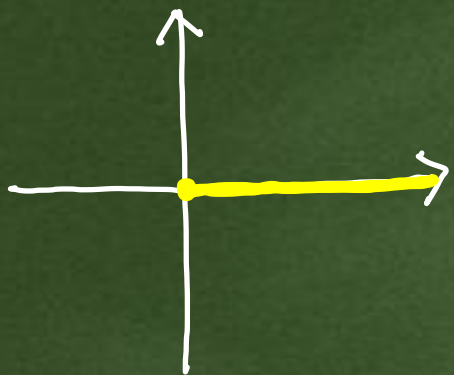


No information about stability / bifurcation,

The conjecture was solved with the aid of the generalized spectral theory (H.C. 2015)

Ex. 2 Laplace op. on $L^2(\mathbb{R}^m)$.

$$\sigma(-\Delta) = \mathbb{R}_{\geq 0}$$



$$A = -\Delta + (\text{perturbation})$$

(e.g. Schrödinger eq.
reaction-diffusion eq.)

$\sigma(A)$ may have nontrivial
(usual) eigenvalues and
generalized eigenvalues.

Generalized e.v. describe ...

- tunnel effect for Schrödinger eq.
- zeros of the Evans function
for RD-eq.

(H.C. 2018)

Ex. 3 The extended theta model (by. K. Kotani)

A model for the onset of brain wave.

V_i : membrain potential of i -th neuron.

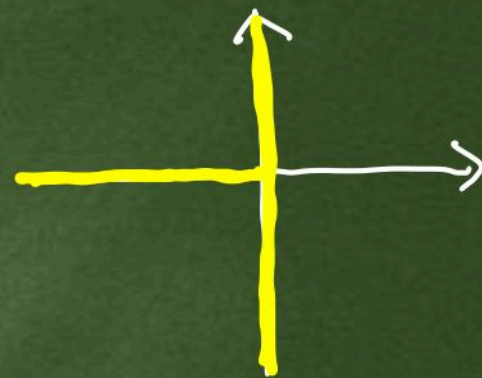
$$\frac{dV_i}{dt} = g_L \frac{(V_i - V_R)(V_i - V_T)}{V_T - V_R} + (\dots)$$

(...) includes interactions and a network structure of neurons.

μ : the number of synapse.

? μ : small $\rightarrow$ no brain wave.
 μ : large $\rightarrow$ stable brain wave.

Linearized op. A
 $\sigma(A)$ is given as



The dynamics is resolved by the generalized spectrum.

(H.C., K. Kotani, A. Akao, 2021)

Ex. 4 Koopman op. / Perron-Frobenius op. of symbolic dyn. systems.

- Full-shift of finite type

$$S: (\omega_1, \omega_2, \dots) \mapsto (\omega_2, \omega_3, \dots)$$

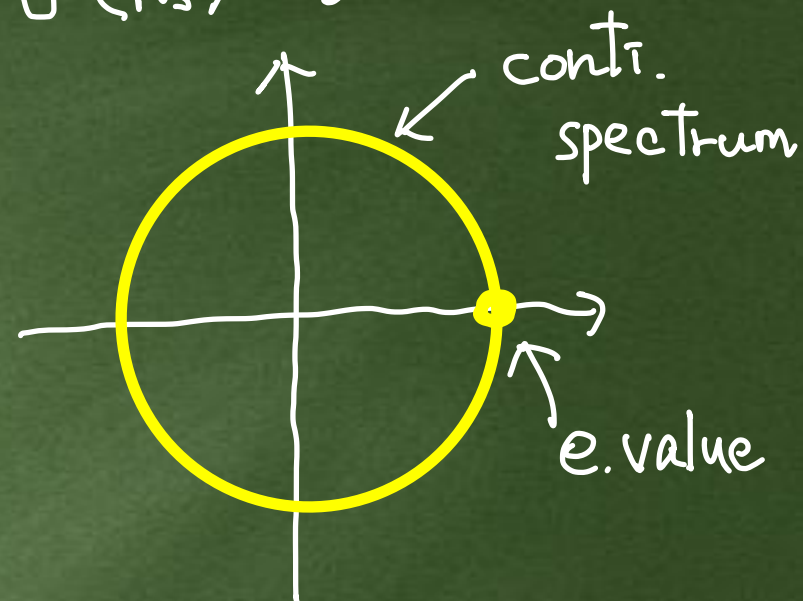
$$\omega_i \in \{0, 1, \dots, p\}, \quad \bar{I} \rightarrow \bar{I}$$

- Koopman op. $K_S: (K_S f)(\omega) := f(S\omega)$

$$L^2(\Sigma) \rightarrow L^2(\bar{I})$$

"chaos" $\Leftrightarrow$ continuous spectrum
on the unit circle.

$\sigma(K_S)$ is



The rate of convergence to chaos is estimated by the generalized spectrum.

(H.C., I. Ishikawa, M. Ikeda)

Generalized Spectral Theory

(H.C. Adv. in. Math, 2015)

$\mathcal{H}$: Hilbert space.

A : Linear operator on $\mathcal{H}$.

$\sigma(A)$: Spectrum

(the set of sing. of $(z - A)^{-1}$)

-
- $X \subset \mathcal{H}$: dense subspace of $\mathcal{H}$
s.t. the topology of X is
stronger than that of $\mathcal{H}$.
 - X' : the dual space of X .

Gelfand triplet

$$X \subset \mathcal{H} \subset X'$$

e.g. $\mathcal{H} = L^2$ and

- $X = C_0^\infty \Rightarrow X' = (\text{Schwarz dist.})$
- $X = (\text{holomorphic functions})$ (Ex. 1.3)
 $\Rightarrow X' = (\text{analytic functionals})$
- $X = L^2(e^{a|x|} dx)$ (Ex. 2)
 $\Rightarrow X' = L^2(e^{-a|x|} dx)$
- $X = \{\text{the set of polynomials}\}$ (Ex. 4) (8)

Gelfand triplet $X \subset H \subset X'$

Thm.

Under "suitable" (technical) assumptions for op. A and X , the resolvent

$$(z - A)^{-1} : X \rightarrow X'$$

the domain and range are controlled

has an analytic continuation beyond the conti. spectrum.

Now the continuous spectrum becomes the branch cut of the (nontrivial) Riemann surface of $(z - A)^{-1} : X \rightarrow X'$

Def.

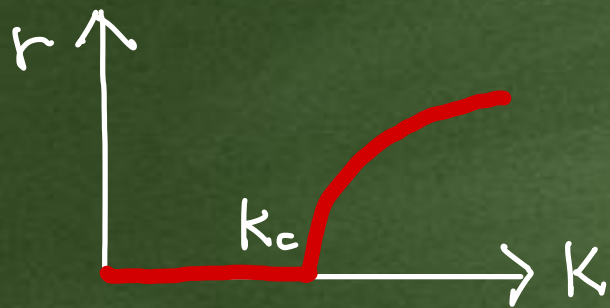
The set of the sing. of the analytic conti. of $(z - A)^{-1}$ on its Riemann surface is called the generalized spectrum.

Rem. The associated eigenvector (if it exists) is in the dual space X' .

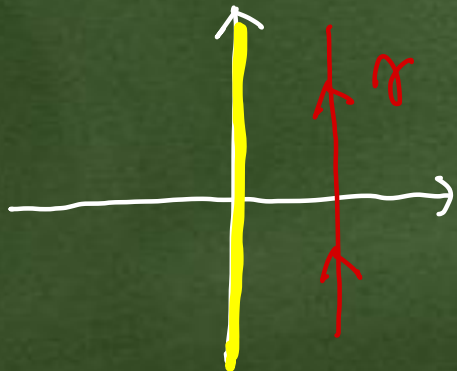
Ex.1 Kuramoto Model (continued)

A : the linearized operator

We want to prove the diagram



For $0 < K < K_c$, $\sigma(A)$ is $i\mathbb{R}$



We want to estimate a solution (semi group) of the linearized eq.

$$d\alpha/dt = A\alpha, \quad \alpha \in L^2$$

given by the Laplace inversion formula

$$e^{At} = \frac{1}{2\pi i} \int_{\gamma} e^{-\lambda t} \underbrace{(\lambda - A)^{-1}}_{\text{resolvent}} d\lambda$$

It is difficult to calculate it because $i\mathbb{R}$ is the set of sing. of $(\lambda - A)^{-1}$.

→ We cannot deform γ toward the left half plane.

Gelfand triplet for A .

$$X \subset L^2 \subset X'$$

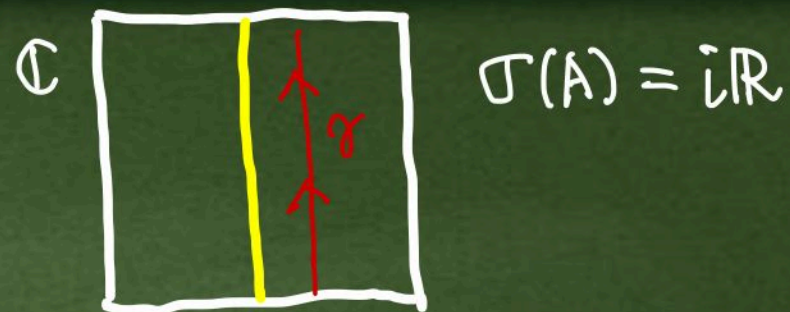
X : some class of holomorphic functions. $\nwarrow$ technical.

Prop.

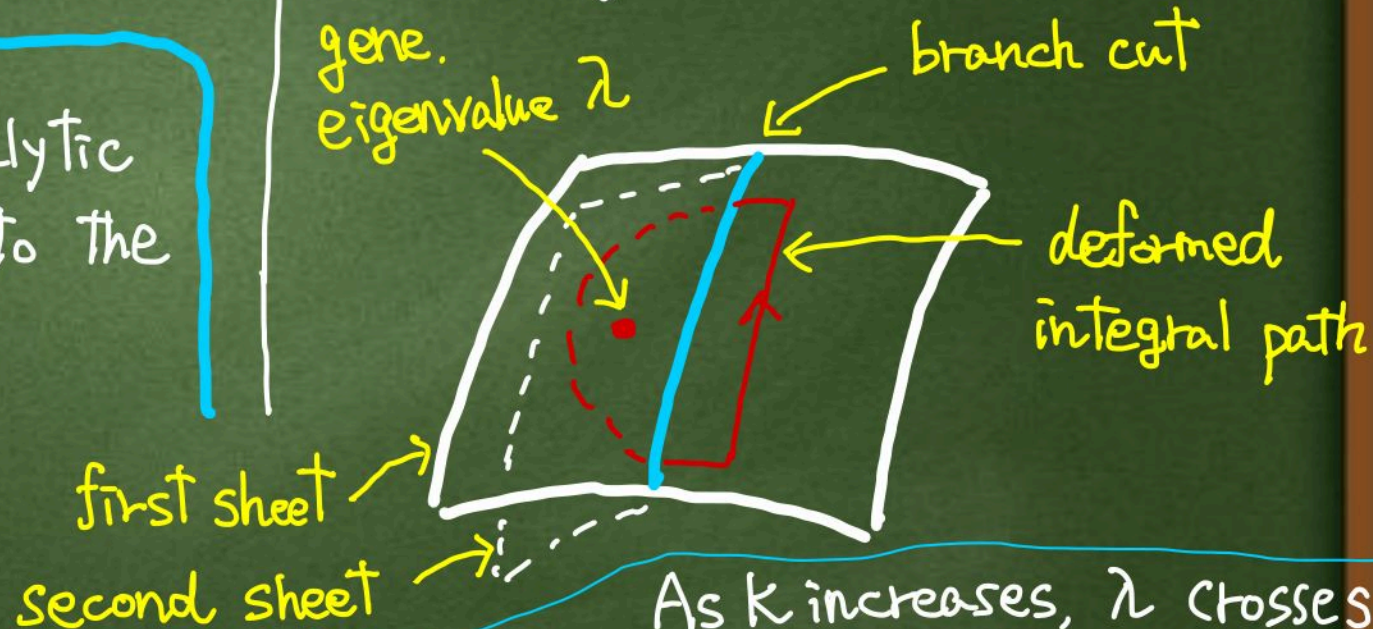
$(\lambda - A)^{-1} : X \rightarrow X'$ has an analytic continuation from the right to the left half plane.

It has a sing. on the second Riemann sheet (gene. e.value).

(H.C. 2015)



$\downarrow$ analy. conti.

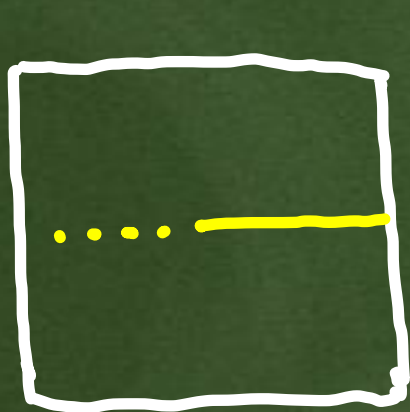
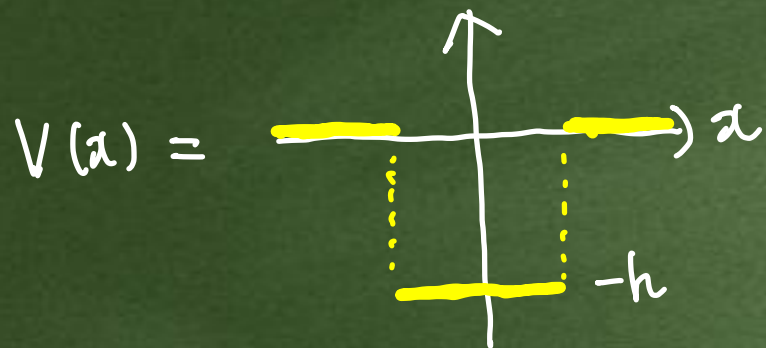


As k increases, λ crosses $i\mathbb{R}$ when $k = k_c$, at which a bifurcation occurs and the conjecture is proved.

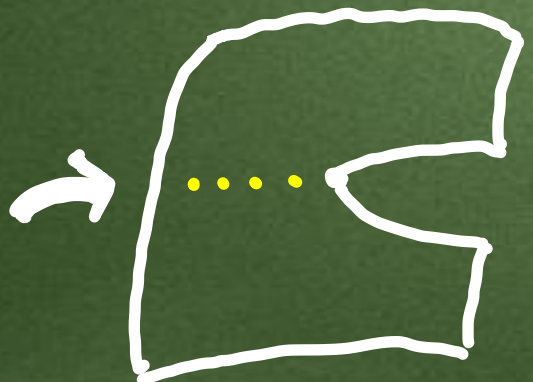
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Ex. 2 Schrödinger eq. (continued)

$$H\psi = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x), \text{ on } \mathbb{R}$$

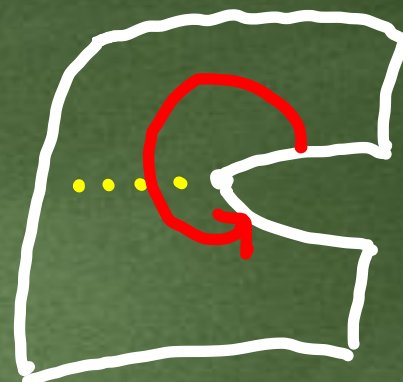


$\sigma(H)$



Cut along the
conti. spectrum

Two copies are glued along the cut
(The Riemann surface of $\sqrt{z}$)



first sheet



second sheet

We find sing of $(\lambda - H)^{-1}$ on the
second sheet $\rightarrow$ generalized eigenvalue.

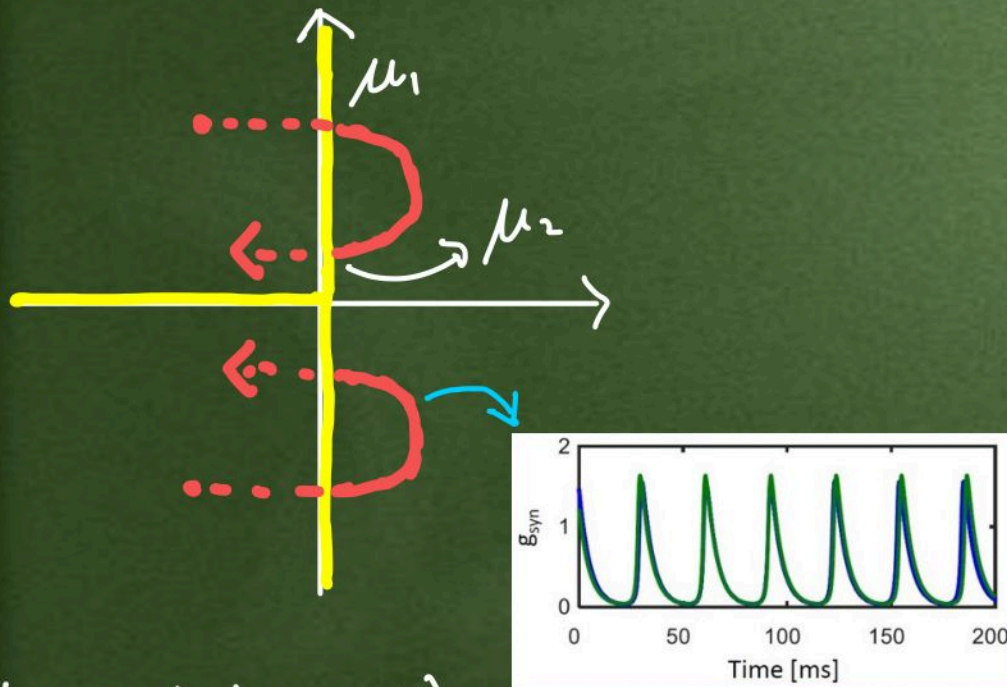
It is related to the time that a particle
remains in the well (tunnel effect).

(H.C. 2018)

Ex3. Brain wave (continued)

A : Linearized op.

μ : Bifurcation parameter
(the number of synapse)



(H.C. et al, 2021)

Red: the motion of the gene. eigenvalues
as μ increases.

Thm. $\exists 0 < \mu_1 < \mu_2$ s.t.

- (i) $0 < \mu < \mu_1$: no brain wave
- (ii) $\mu = \mu_1$: Hopf bifurcation
- (iii) $\mu_1 < \mu < \mu_2$: $\exists$ stable Gamma wave
30 ~ 80 Hz
- (iv) $\mu = \mu_2$: again Hopf
- (v) $\mu_2 < \mu$: no brain wave

The gamma wave exists only for
intermediate values μ

→ Synaptic pruning in early childhood? (13)

Ex. 4 Symbolic dynamical systems (continued)

Shift op. $S: (\omega_1, \omega_2, \dots) \mapsto (\omega_2, \omega_3, \dots)$

Koopman op. $K_S: (K_S f)(\omega) = f(S\omega)$

$$X \subset L^2(\Sigma) \subset X',$$

X : the space of polynomials on Σ

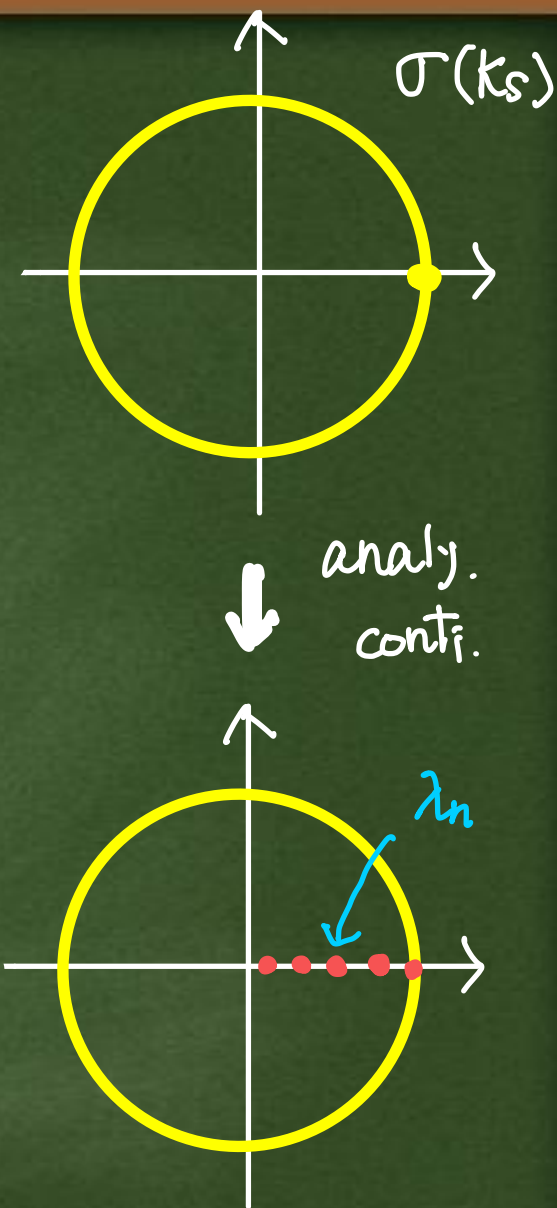
$\exists$ Analytic conti. from the outside to the inside of the unit circle.
of $(\lambda - K_S)^{-1}$

Thm.

Gene. eigenvalues are given by $\lambda_n = 1/2^n$ ($n=0, 1, \dots$).
The spectral decomposition holds:

$$K_S \mu = \sum_{n=0}^{\infty} \lambda_n \langle \mu | \bar{\Phi}_n \rangle \bar{\Phi}_n', \quad \forall \mu \in X'$$

(H.C. et al. 2023)



Thank you!!

